



Chapter 3

氧化工艺

Oxidation of Silicon

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2025年11月17日



- ✎ **3. 1 Why is oxidation needed?**
- ✎ **3. 2 Basics of thermal oxidation**
- ✎ **3. 3 Kinetics of thermal oxidation**
- ✎ **3. 4 Other aspects of thermal oxidation**
- ✎ **3. 5 Summary**

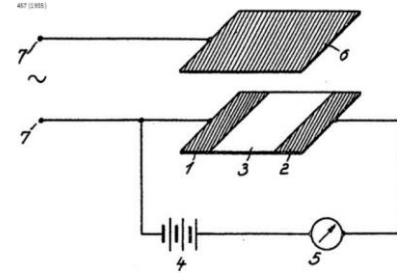
3. 1 Why is oxidation of Si needed?

Ge has high μ_e , μ_h ,
Ge is stable...,
...**but no good oxide**

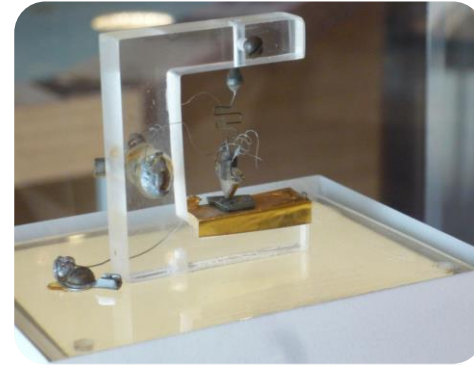
GaAs has high μ_e
and direct band...,
...**but no good oxide**

Probably safe to say that
history of IC would be
different without good
 SiO_2 grown on Si wafer

Concept of FET, patent
(1925)
(1935)



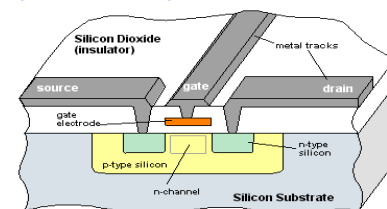
Realization of FET
(1947)



Realization of MOSFET
(1959)

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NMOS Transistor
(n-channel MOSFET)

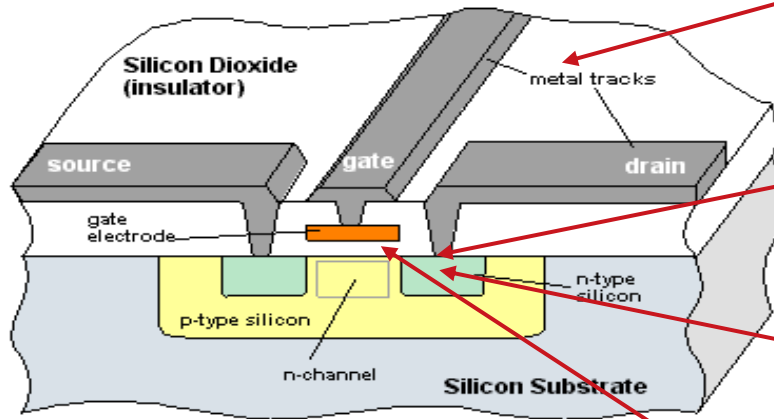


3. 1 Why is oxidation of Si needed?

MOSFET

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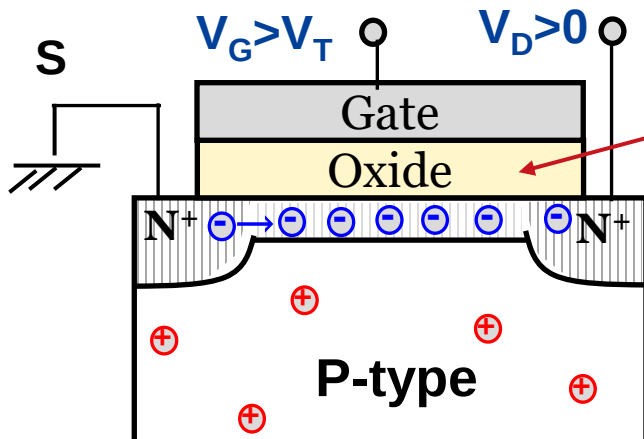
NMOS Transistor (n-channel MOSFET)



Insulator is needed between electrodes and devices

Si surface should be protected during implantation

Protected surface should be cleaned for some process

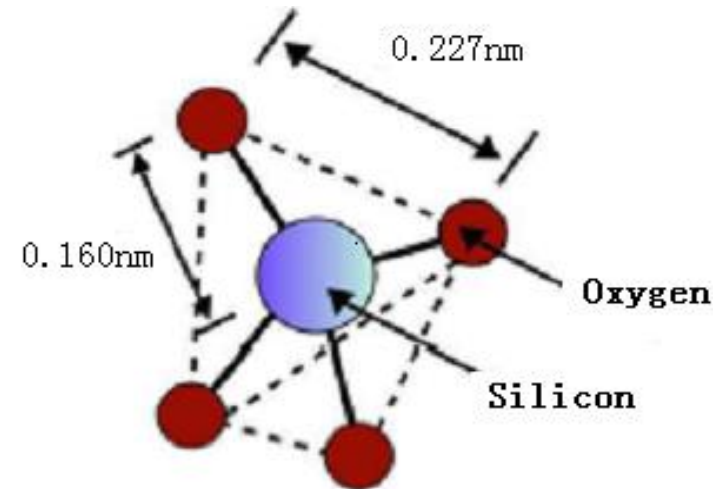


MOS need very very very good oxide and oxide/semiconductor interface

IC should be protected from machinal and chemical damaging.

3. 1 Why is oxidation of Si needed?

- SiO_2 is stable down to 10^{-9} Torr, $T > 900^\circ\text{C}$
- SiO_2 is a diffusion barrier for B, P, As
- SiO_2 can be etched with HF which leaves Si unaffected
- SiO_2 is good insulator, $E_g > 8\text{eV}$
- SiO_2 has high dielectric breakdown field, $500\text{V}/\mu\text{m}$
- SiO_2 grown on Si $\rightarrow$ clean Si/ SiO_2



SiO_2 是上帝赐给集成电路的材料 (天选之材)

- ✎ **3. 1 Why is oxidation needed?**
- ✎ **3. 2 Basics of thermal oxidation**
- ✎ **3. 3 Kinetics of thermal oxidation**
- ✎ **3. 4 Other aspects of thermal oxidation**
- ✎ **3. 5 Summary**

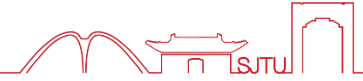
3. 2 Why thermal oxidation is needed?



How can SiO_2 be prepared on Si ?

- Chemical vapor deposition (CVD)
- Physical vapor deposition
- Anodic oxidation
- **Thermal oxidation (best quality)**

3.2.1 Properties of thermal SiO₂

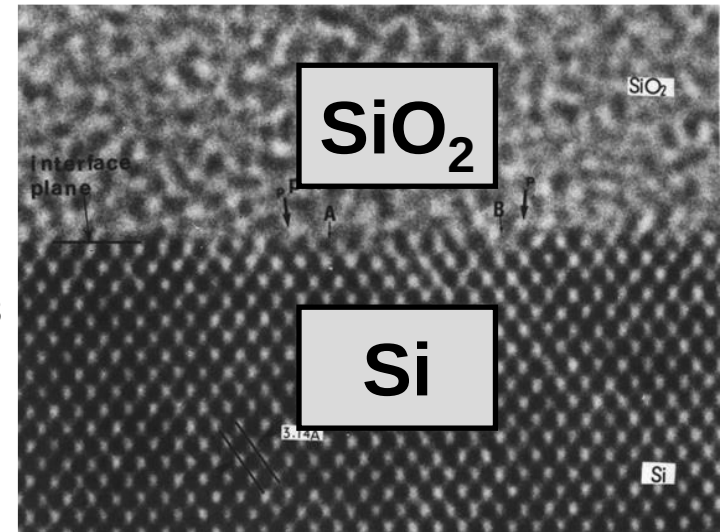


Thermal SiO₂ is amorphous.

Weight Density = 2.20 gm/cm³

Molecular Density = 2.3E22 molecules/cm³

Crystalline SiO₂ (Quartz) = 2.65 gm/cm³



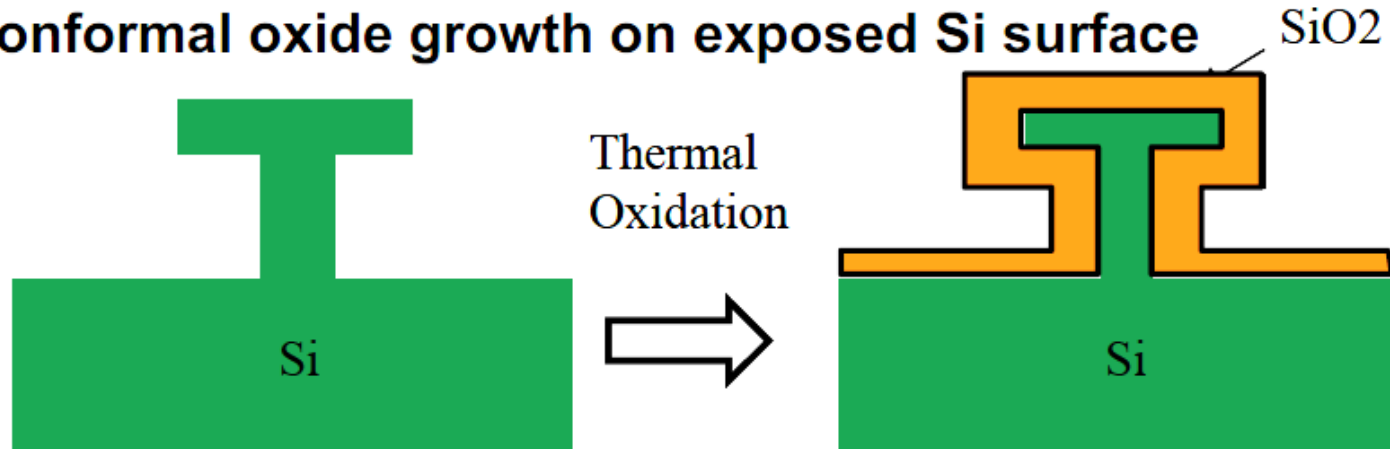
(1) Excellent Electrical Insulator

Resistivity > 10²⁰ ohm--cm Energy Gap ~ 9 eV

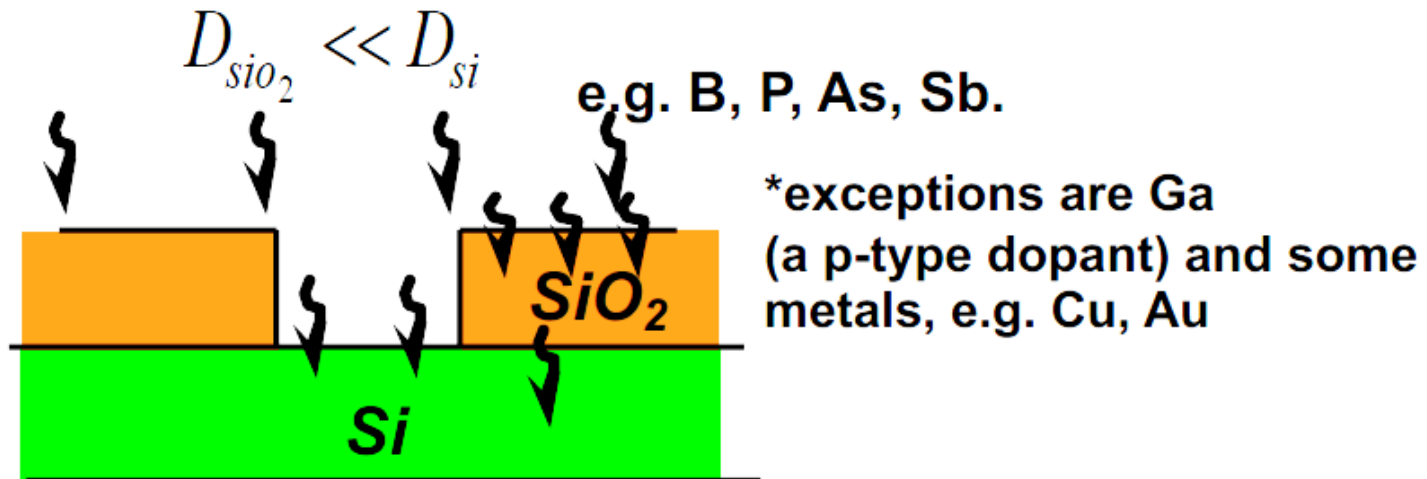
(2) High Breakdown Electric Field > 10MV/cm

(3) Stable and Reproducible Si/SiO₂ Interface

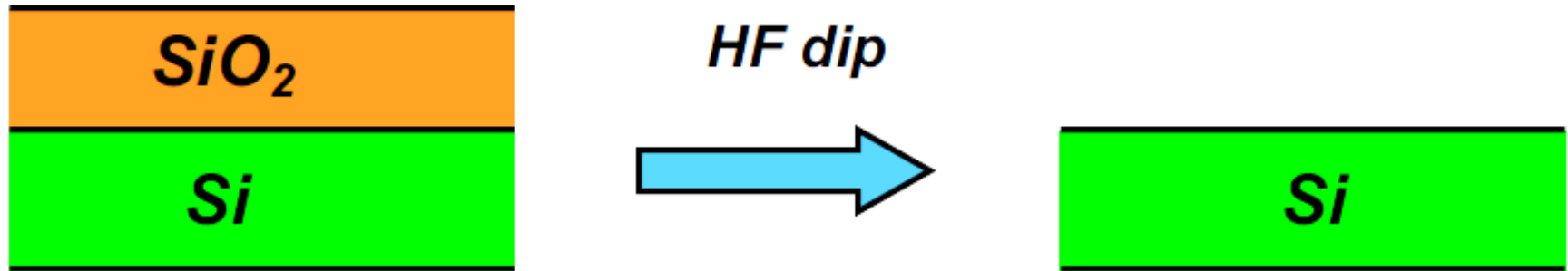
(4) Conformal oxide growth on exposed Si surface



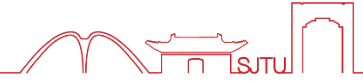
(5) SiO₂ is a good diffusion mask for common dopants



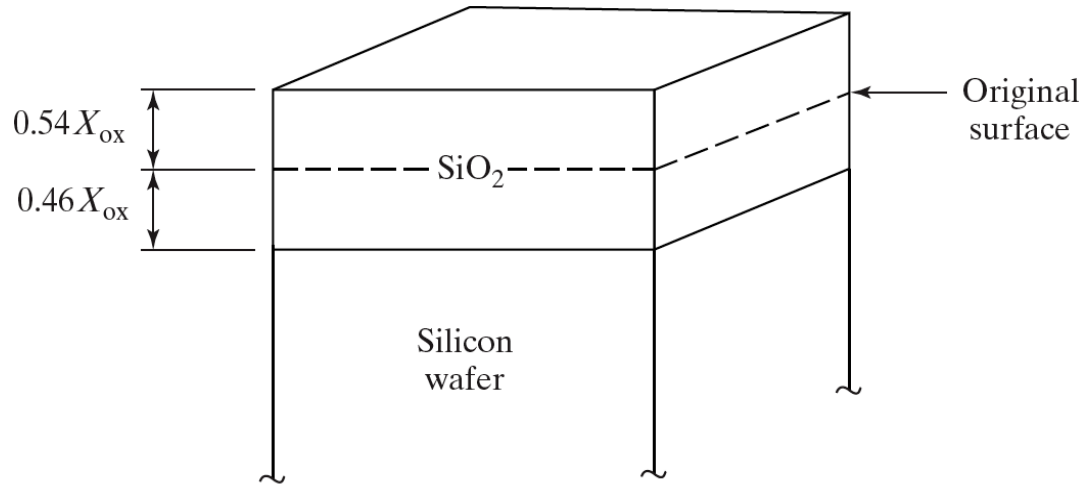
(6) Very good etching selectivity between Si and SiO_2 .



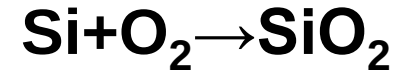
3.2.2 Method of thermal oxidation



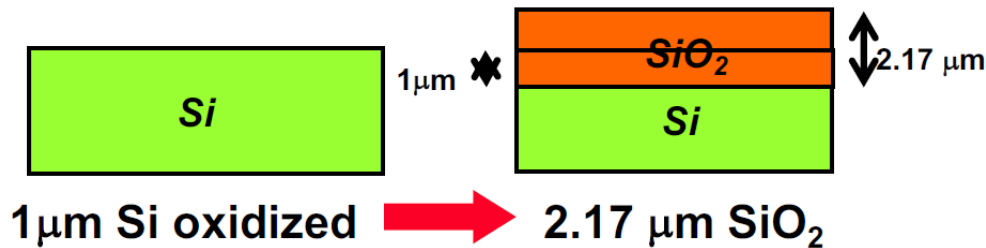
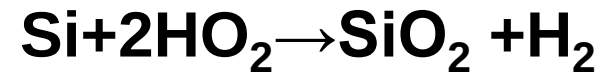
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Dry oxidation



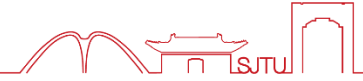
Wet oxidation



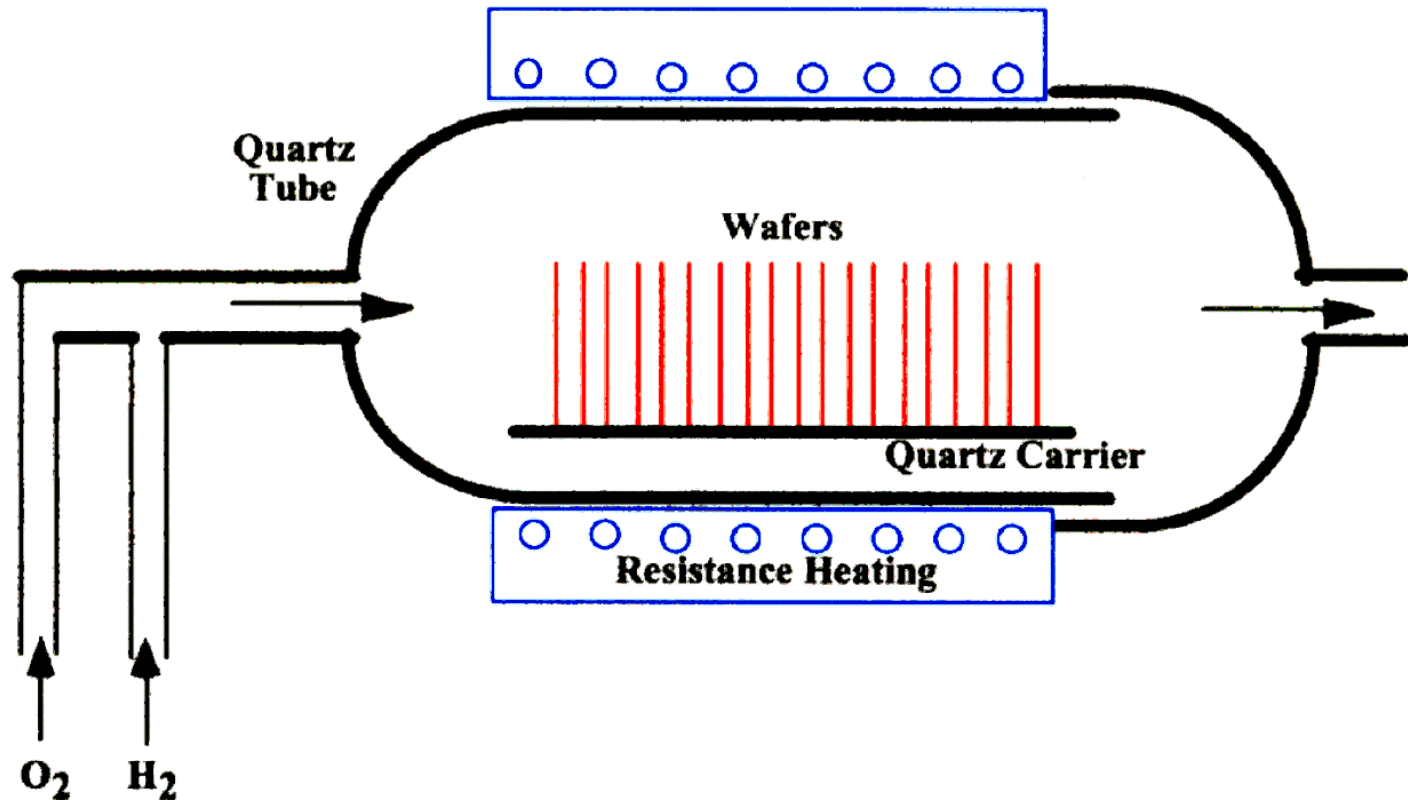
$$X_{si} = X_{ox} \cdot \frac{N_{ox}}{N_{si}}$$

Growth Occurs 54% above and 46% below original surface as silicon is consumed.

3.2.2 Facilities of thermal oxidation



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- Oxidation systems are conceptually very simple
- In practice today, vertical furnaces, rapid thermal oxidation (RTO) systems, and fast ramp furnaces are in use

3.2.2 Facilities of thermal oxidation



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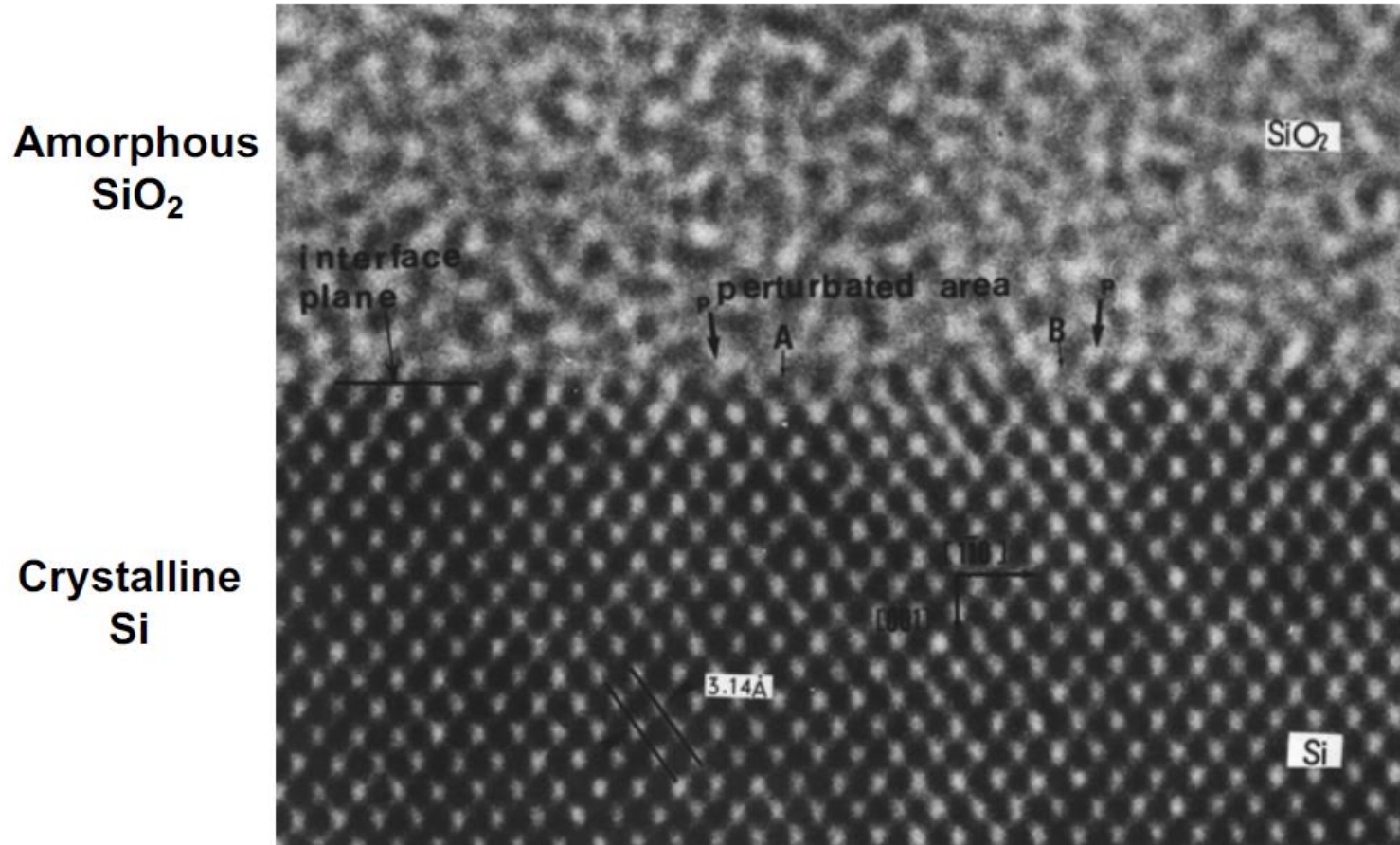


Horizontal Furnace



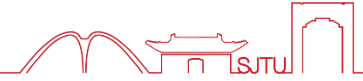
Vertical Furnace

Transmission Electron Micrograph

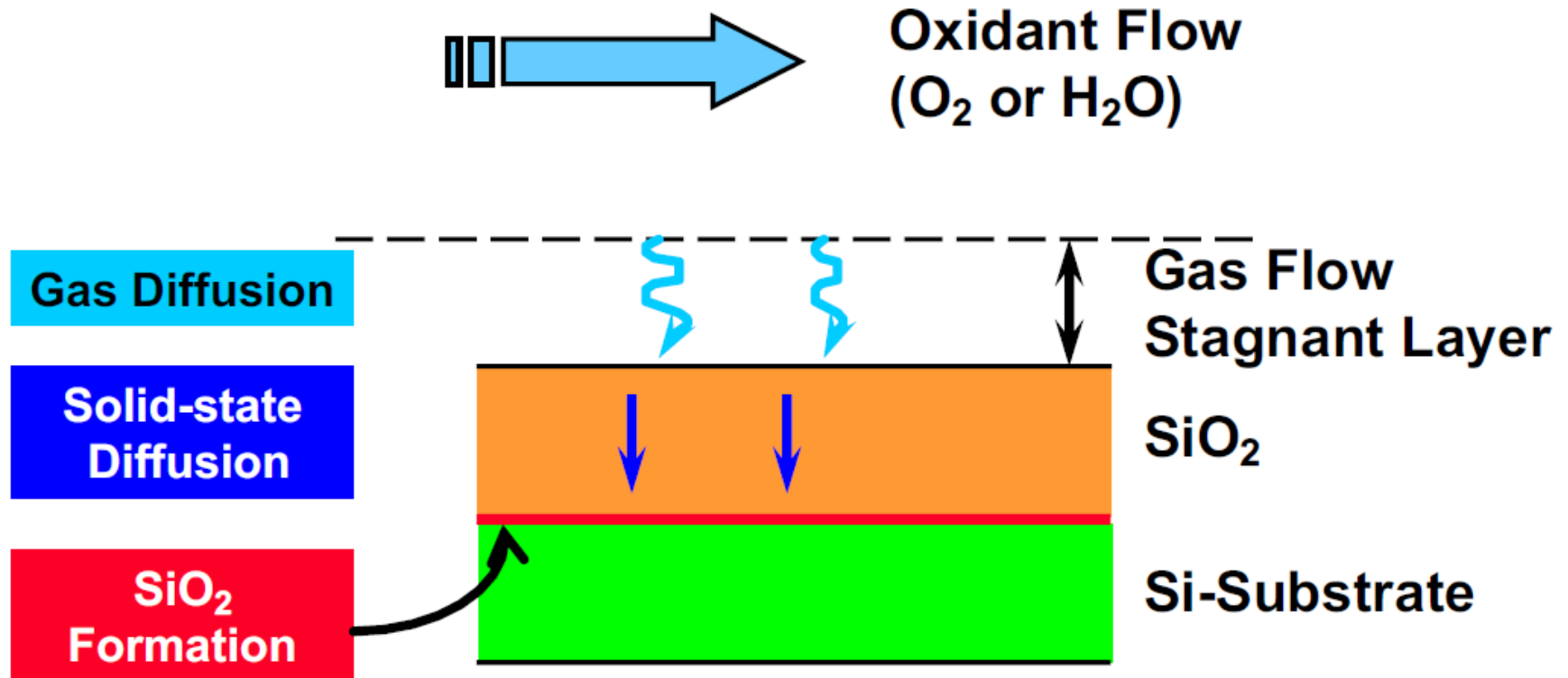


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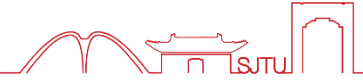
3.3.1 Kinetics of thermal oxidation



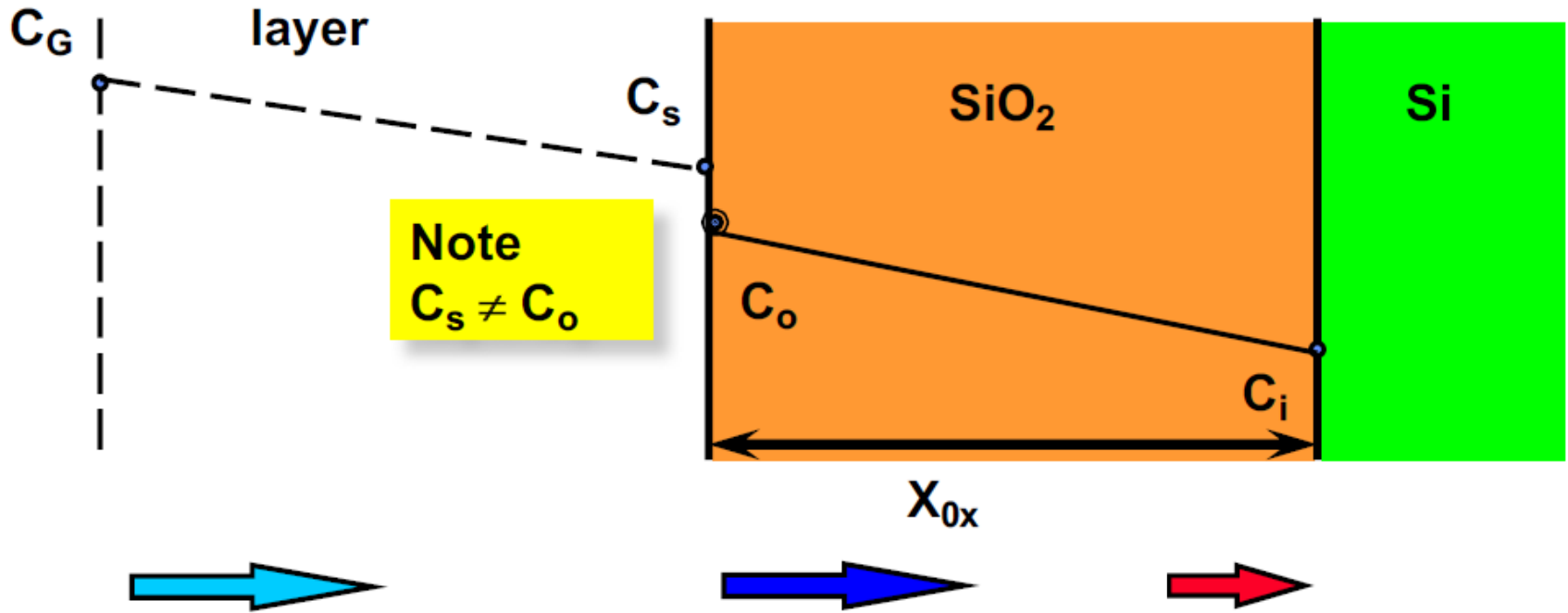
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3.3.2 Deal-Grove Model



B. E. Deal and A. S. Grove, J. Appl. Phys. 36, 3771 (1965).



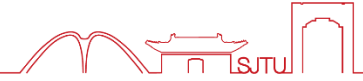
Gas transport

Diffusion through SiO_2

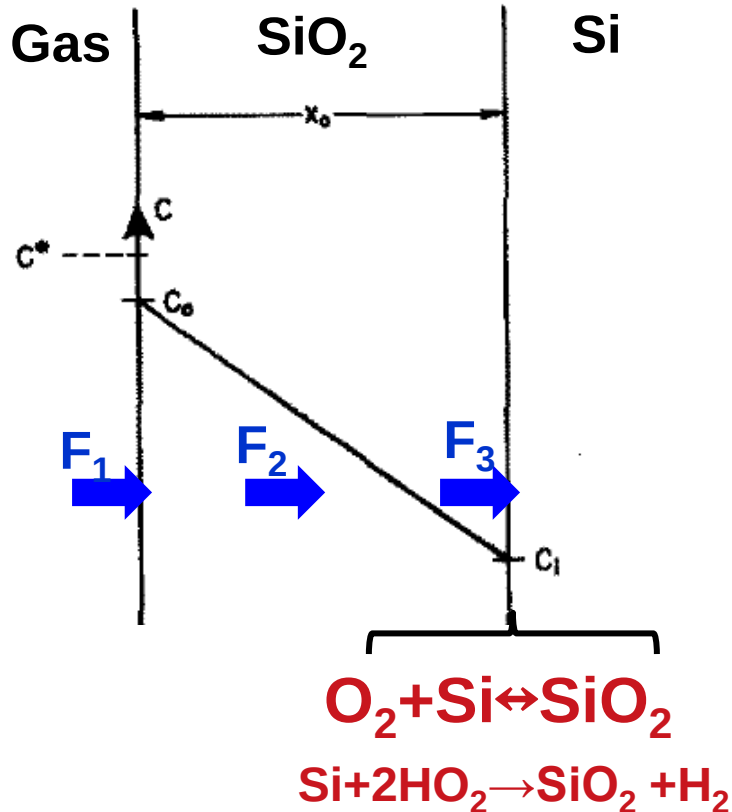
Reaction at interface

Objective: formulate $x=f(t)$

3.3.2 Deal-Grove Model-modeling



F: flux – the number of oxygen molecules that crosses a plane per unit area per second



At surface

$$F_1 = h(C^* - C_o) \quad h_g \text{ 是质量输运系数}$$

h: Mass transfer coefficient [cm/sec].

In SiO₂ D_{eff}: Diffusivity [cm²/sec]

$$F_2 = D_{eff} \frac{(C_o - C_i)}{x} \quad \text{菲克第一定律}$$

At SiO₂/Si interface

$$F_3 = kC_i \quad \text{k: Oxidation reaction rate constant}$$

Oxidant diffusion → interface reaction.

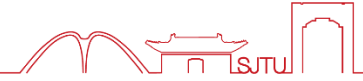
In SiO_2 is growth, the Si at the interface is changed to SiO_2 . Thus, the flux at the interface is :

Oxidation rate

$$-\frac{dx}{dt} = \frac{F}{N_1}$$

N_1 : oxidant molecules/unit volume required to form a unit volume of SiO_2 . The molecular density of SiO_2 is $2.2 \times 10^{22}/\text{cm}^3$. In oxidation, to get a SiO_2 molecular, we need a O_2 molecular in dry oxidation or two H_2O moleculars in wet oxidation.

3.3.2 Deal-Grove Model-modeling



C^* is a controlled process variable proportional to the O_2 pressure

Only C_o and C_i are the 2 unknown variables

At steady state

$$F_1 = F_2 = F_3$$

$$C_i = \frac{C^*}{1 + \frac{k}{h} + \frac{kx}{D_{eff}}}$$

$$C_o = \frac{(1 + \frac{kx}{D_{eff}})C^*}{1 + \frac{k}{h} + \frac{kx}{D_{eff}}}$$

■ When $D_{eff} \ll kx$,

$$C_i \sim 0, C_o \sim C^*,$$

oxidation is mainly controlled by diffusion.

■ When D_{eff} is very large, $C_i = C_o = C^*/(1+k/h)$,

oxidation is mainly controlled by reaction.

In SiO_2 is growth, the Si at the interface is changed to SiO_2 . Thus, the flux at the interface is :

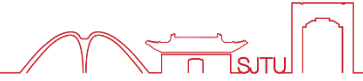
Oxidation rate

$$-\frac{dx}{dt} = \frac{F}{N_1}$$

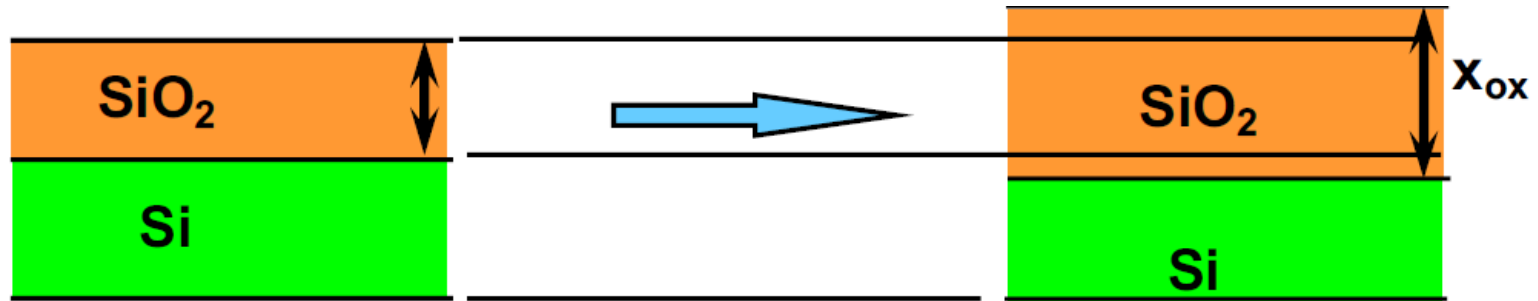
$$F = N_1 \frac{dx}{dt} = kC_i = k \frac{C^*}{1 + \frac{k}{h} + \frac{kx}{D_{eff}}}$$

N_1 : oxidant molecules/unit volume required to form a unit volume of SiO_2 . The molecular density of SiO_2 is $2.2 \times 10^{22}/\text{cm}^3$. In oxidation, to get a SiO_2 molecular, we need a O_2 molecular in dry oxidation or two H_2O moleculars in wet oxidation.

3.3.2 Deal-Grove Model-solution



Initial Condition: At $t = 0$, $X = X_0$



$$x_0^2 - x^2 + A(x_0 - x) = Bt$$

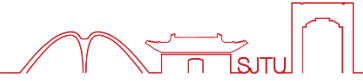
$$x^2 + Ax = B(t + \tau)$$

$$A \equiv 2D_{eff} \left(\frac{1}{k} + \frac{1}{h} \right)$$

$$B \equiv 2D_{eff} \frac{C^*}{N_1}$$

$$\tau = x_0^2 + Ax_0$$

3.3.2 Deal-Grove Model-modeling



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Only C_o and C_i are the 2 unknown variables

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$$F_1 = F_2 = F_3$$

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■ When $D_{eff} \ll kx$,

$$C_i \sim 0, C_o \sim C^*,$$

oxidation is mainly controlled by diffusion.

■ When D_{eff} is very large, $C_i =$

$$C_o = C^* / (1 + k/h),$$

oxidation is mainly controlled by reaction.

$$x, = \frac{A}{2} \left(\sqrt{1 + \frac{t + \tau}{A^2/4B}} - 1 \right)$$

When $t \ll A^2/4B$ 时,

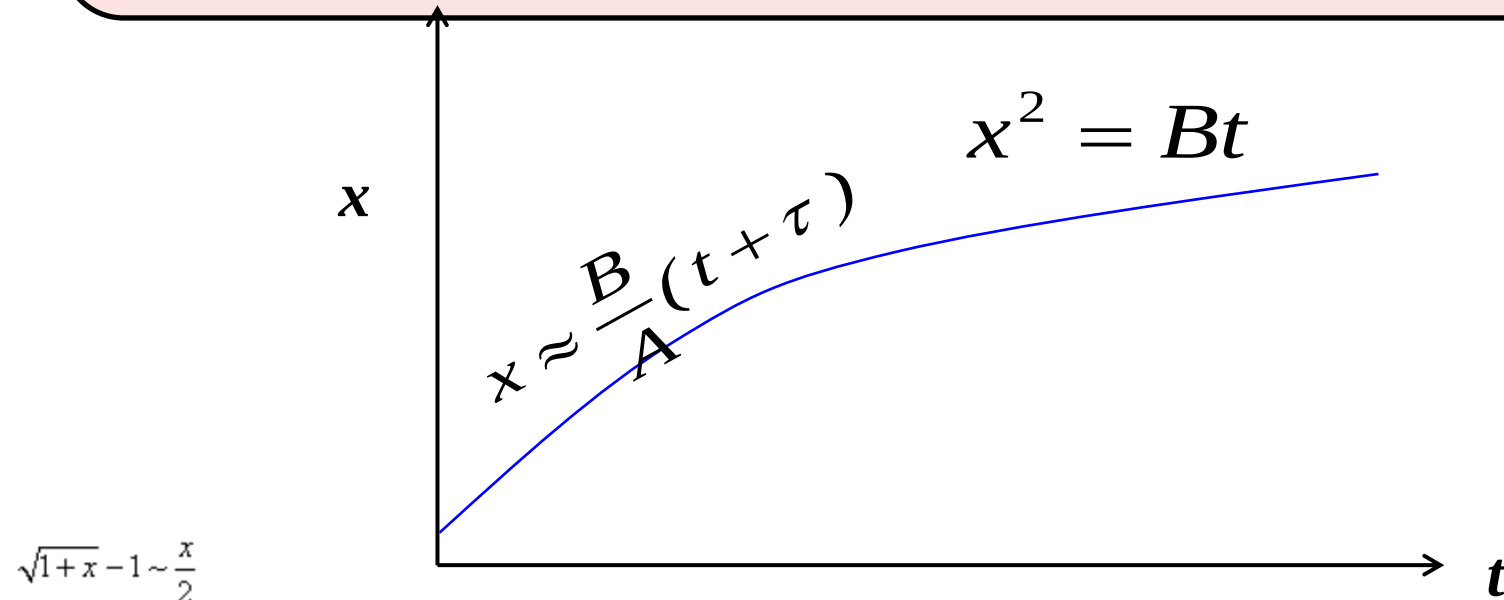
$$x \approx \frac{B}{A}(t + \tau)$$

(线性规律, 反应控制)

When $t \gg A^2/4B$,

$$x^2 \approx B(t + \tau)$$

(抛物型规律, 扩散控制)



Rate constants B and B/A have physical meaning:
oxidant diffusion and interface reaction rate respectively

$$B = C_1 \exp(-E_1 / kT)$$

$$\frac{B}{A} = C_2 \exp(-E_2 / kT)$$

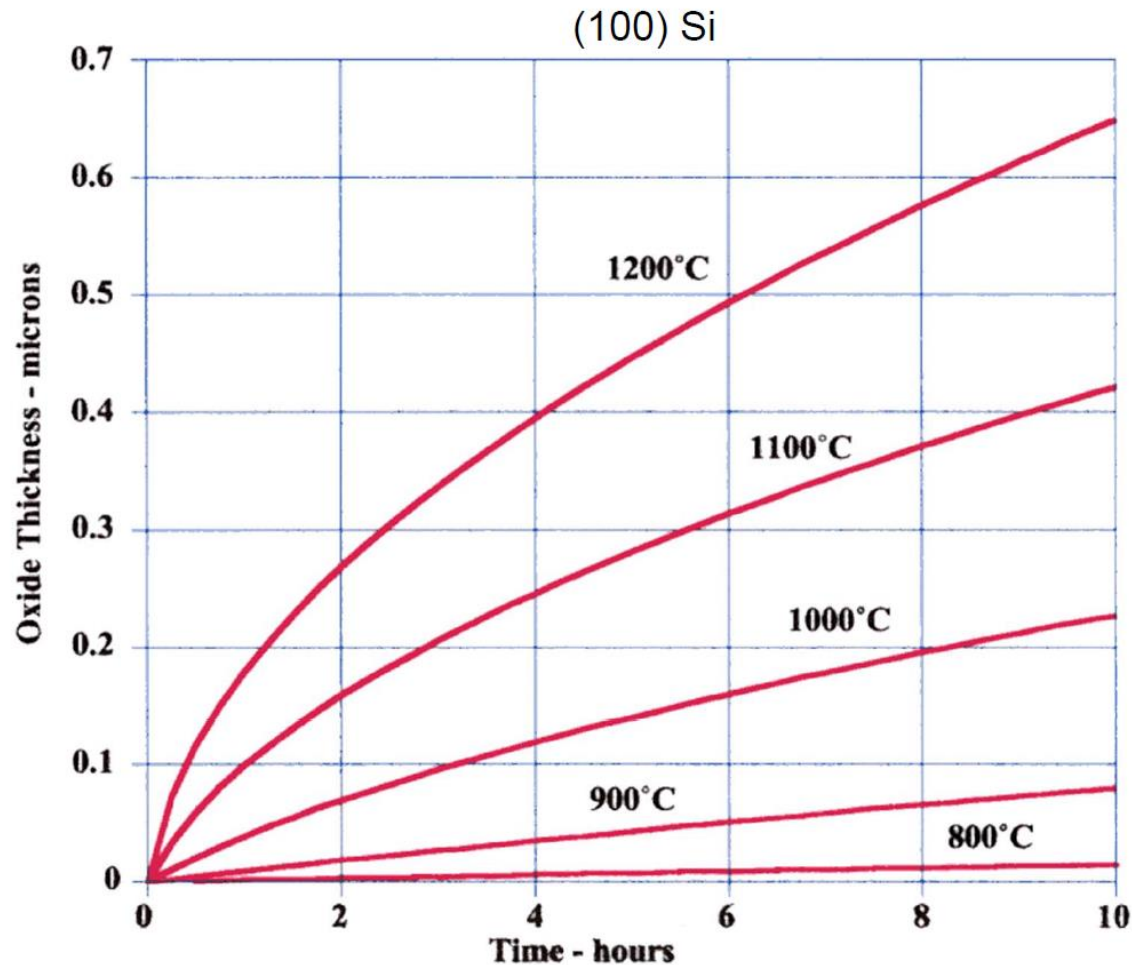
$$B \equiv 2D_{SiO_2} \frac{C^*}{N_1}$$

$$A \equiv 2D_{SiO_2} \left(\frac{1}{k} + \frac{1}{h} \right)$$

Ambient	B	B/A
Dry O ₂	$C_1 = 7.72 \times 10^2 \mu^2 \text{ hr}^{-1}$ $E_1 = 1.23 \text{ eV}$	$C_2 = 6.23 \times 10^6 \mu \text{ hr}^{-1}$ $E_2 = 2.0 \text{ eV}$
Wet O ₂	$C_1 = 2.14 \times 10^2 \mu^2 \text{ hr}^{-1}$ $E_1 = 0.71 \text{ eV}$	$C_2 = 8.95 \times 10^7 \mu \text{ hr}^{-1}$ $E_2 = 2.05 \text{ eV}$

- activation energy for B varies with process (diffusivity of different species through the oxide)
- activation energy for B/A is roughly constant (surface reaction rate)

3.3.2 Deal-Grove Model-simulation



- **Calculated dry O_2 oxidation rates using Deal Grove.**

- linear regime followed by parabolic behavior
- dry oxidations typically used for oxide thicknesses < 100 to 200 nm

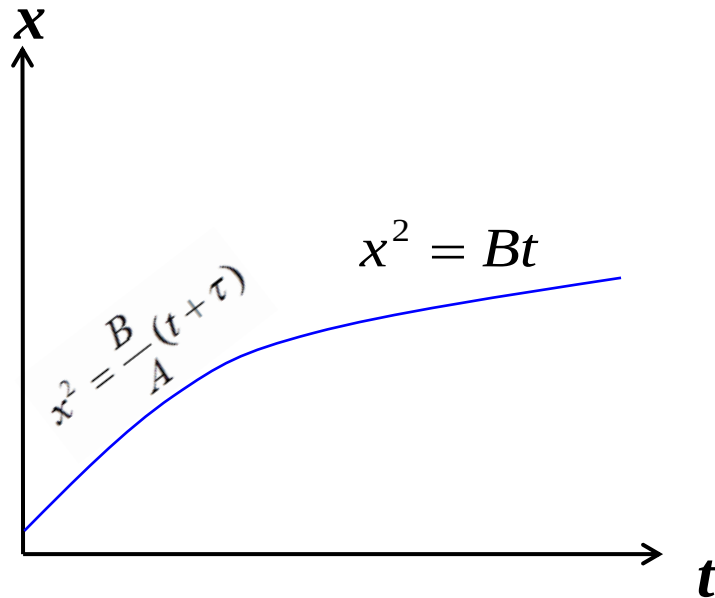


- Partial pressure of oxidant (氧化剂分压)
- Temperature of oxidation (氧化温度)
- Surface orientation of Si (硅晶面取向)
- Doping and Impurity (杂质)
- Dry and Wet oxidation (干湿法氧化)

3.3.3 Factors influencing oxidation rate (1)

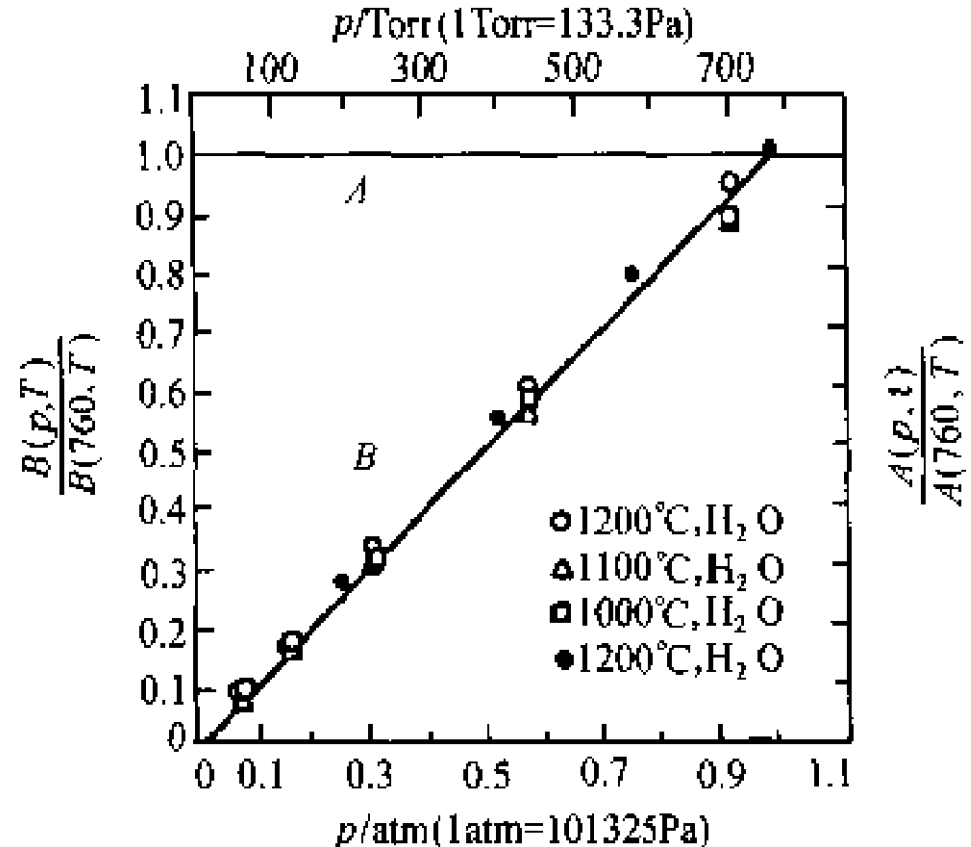
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Effect of Partial pressure of oxidant (氧化剂分压影响)



$$B \equiv 2D_{SiO_2} \frac{C^*}{N_1}$$

$$C^* = Hp_g$$



$P \uparrow, C^* \uparrow \Rightarrow B \uparrow$

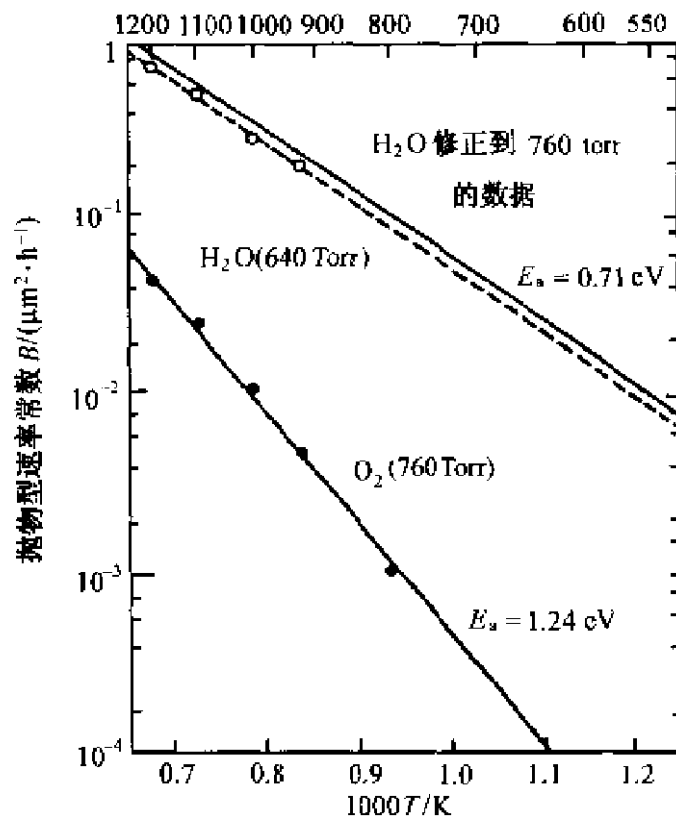
3.3.3 Factors influencing oxidation rate (2)

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Effect of temperature of oxidant (温度影响)

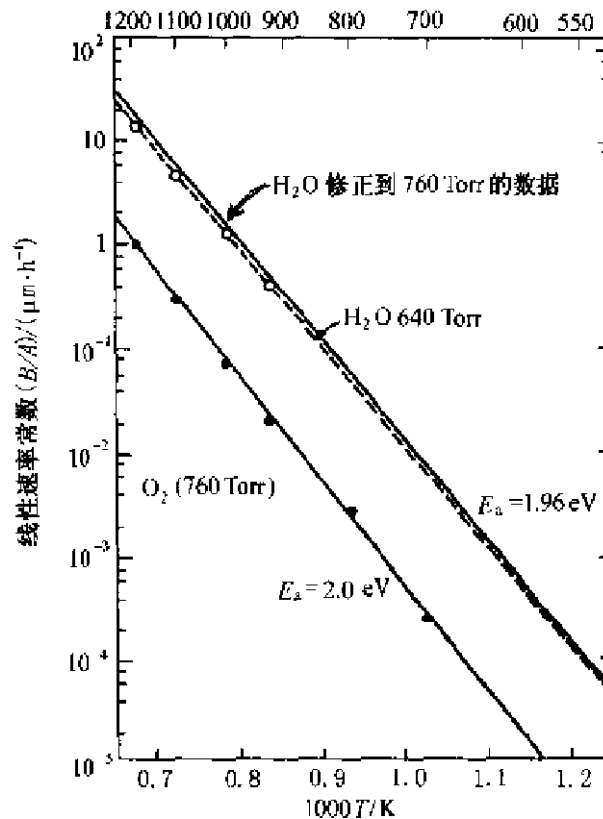
$$B \equiv 2D_{SiO_2} \frac{C^*}{N_1}$$

温度/ $^{\circ}\text{C}$



$$A \equiv 2D_{SiO_2} \left(\frac{1}{k} + \frac{1}{h} \right)$$

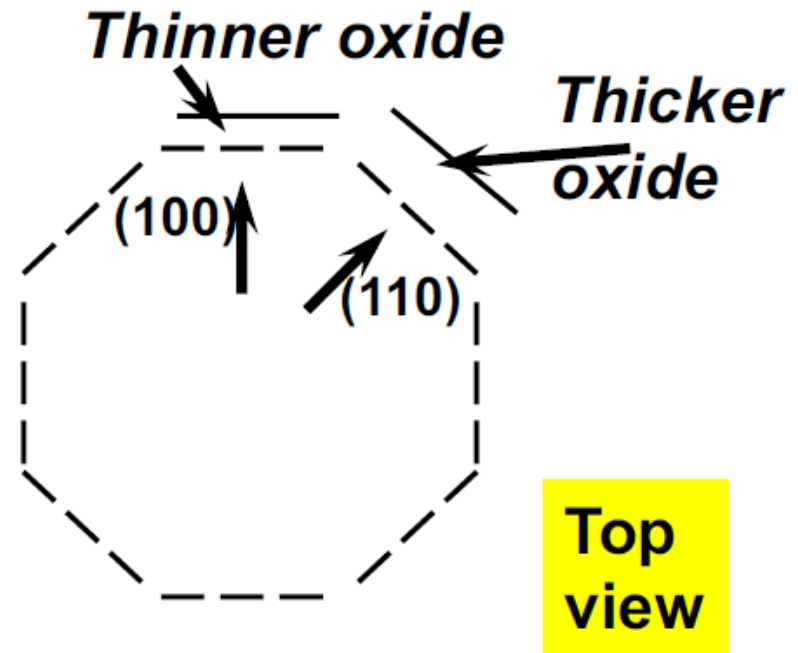
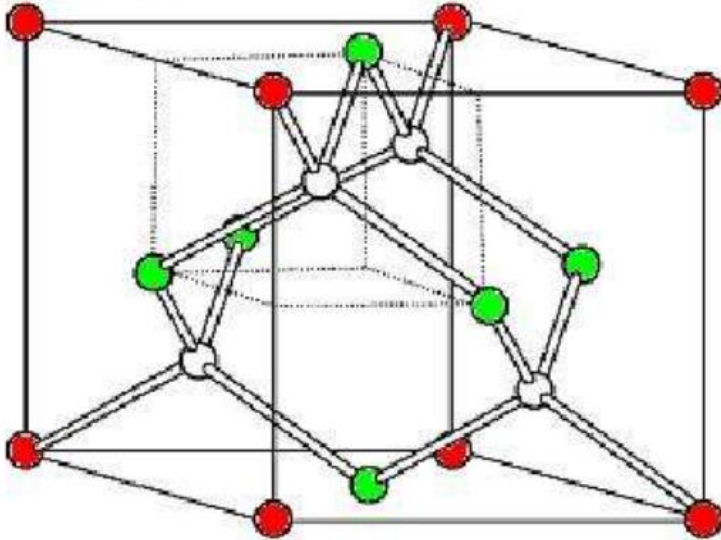
温度 $T/^{\circ}\text{C}$



$T \uparrow, D_{SiO_2} \uparrow, K_s \uparrow \Rightarrow B \uparrow, B/A \uparrow$

3.3.3 Factors influencing oxidation rate (3)

Effect of surface orientation (硅晶体取向影响)

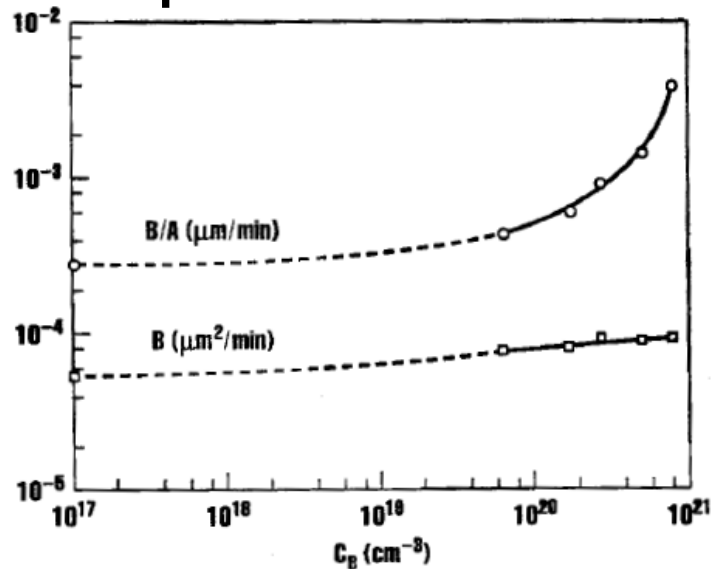


$$A \equiv 2D_{SiO_2} \left(\frac{1}{\boxed{k}} + \frac{1}{h} \right)$$

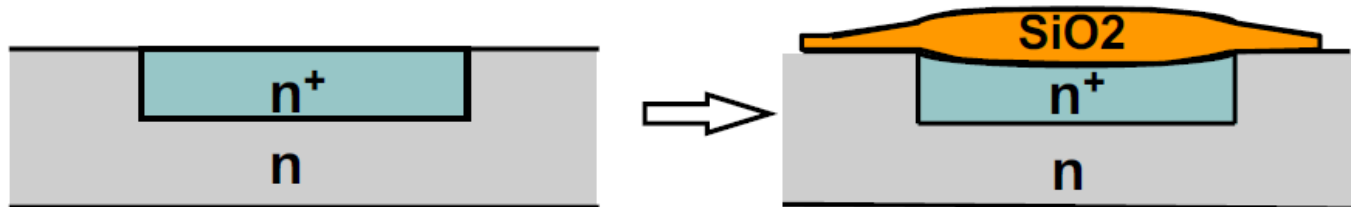
$$\underline{(111) > (110) > (100)}$$

Effect of doping (掺杂浓度影响)

Coefficients for dry oxidation at 900°C as function of surface Phosphorus concentration



Dry oxidation, 900°C

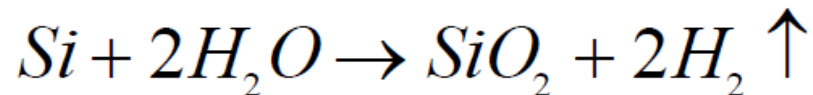


Dry / Wet Oxidation

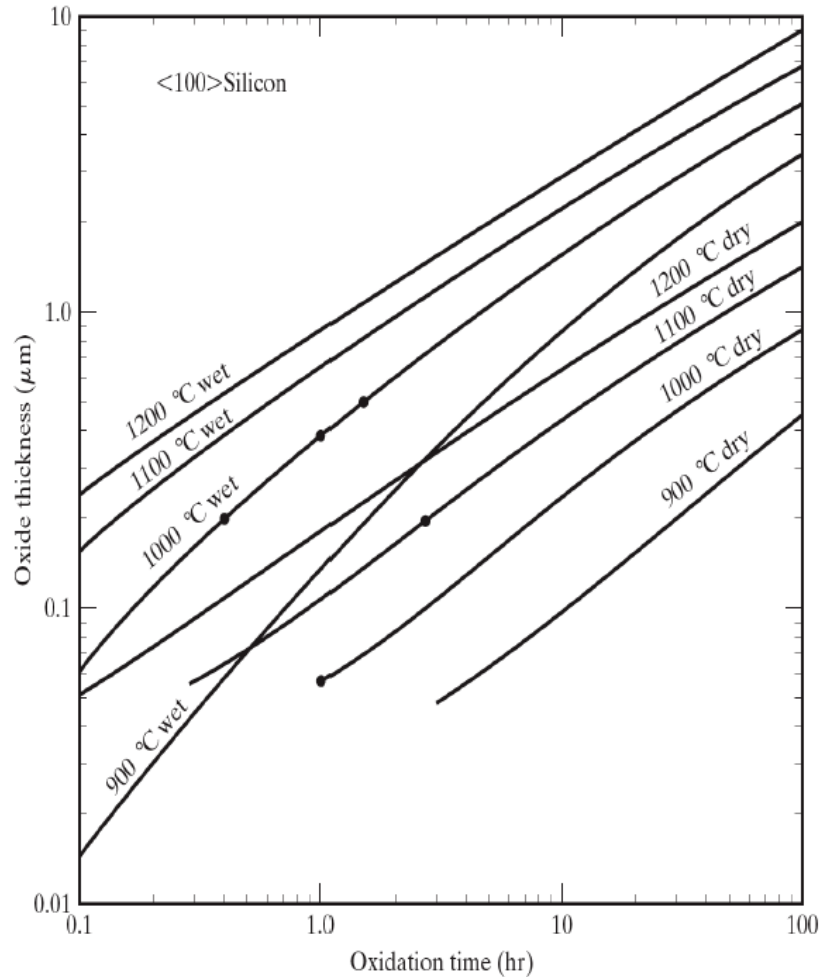
Note : “dry” and “wet” oxidation have different N1 factors

$$N_1 = 2.3 \times 10^{22} / \text{cm}^3 \quad \text{for } O_2 \text{ as oxidant}$$
$$Si + O_2 \rightarrow SiO_2$$

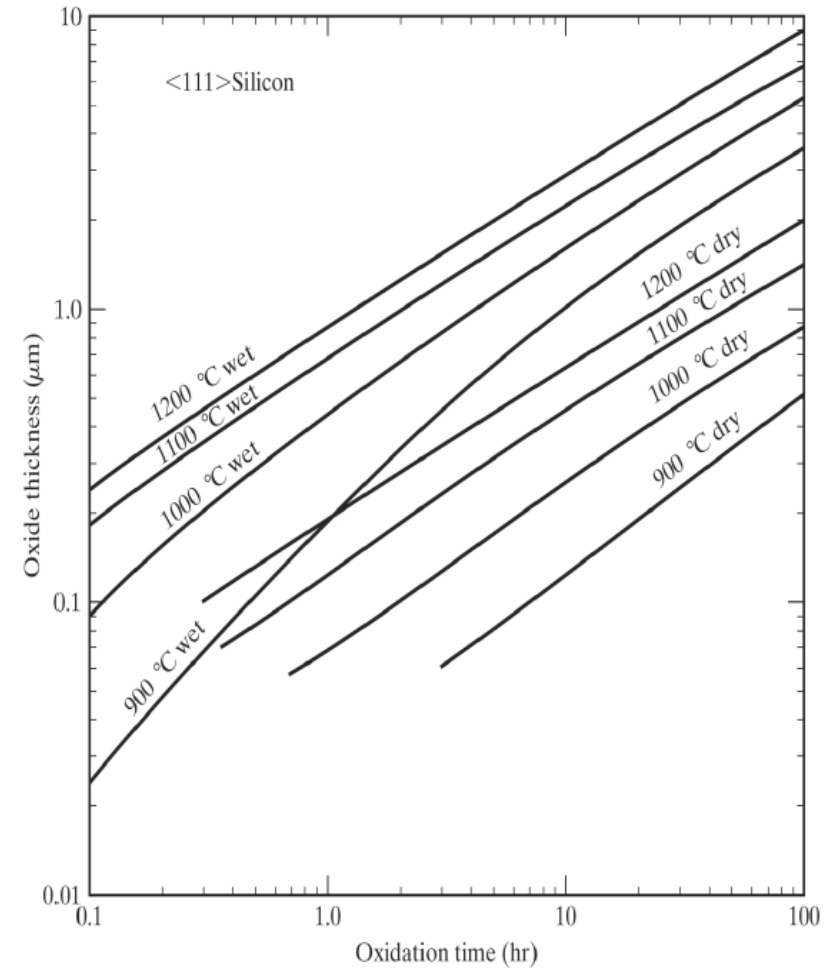
$$N_1 = 4.6 \times 10^{22} / \text{cm}^3 \quad \text{for } H_2O \text{ as oxidant}$$



Thermal Oxidation on <100> Silicon



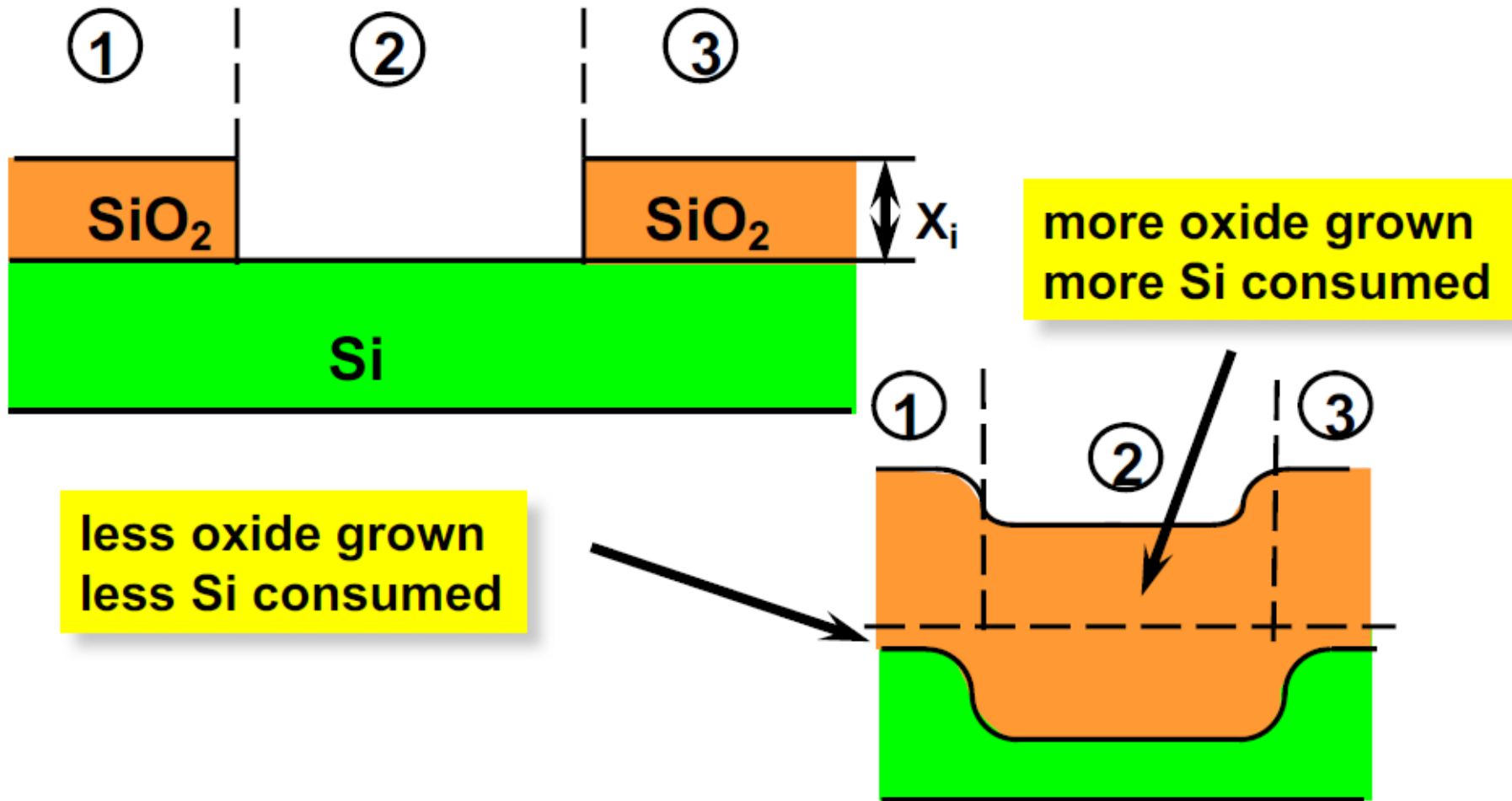
Thermal Oxidation on <111> Silicon



3.3.3 Factors influencing oxidation rate (6)

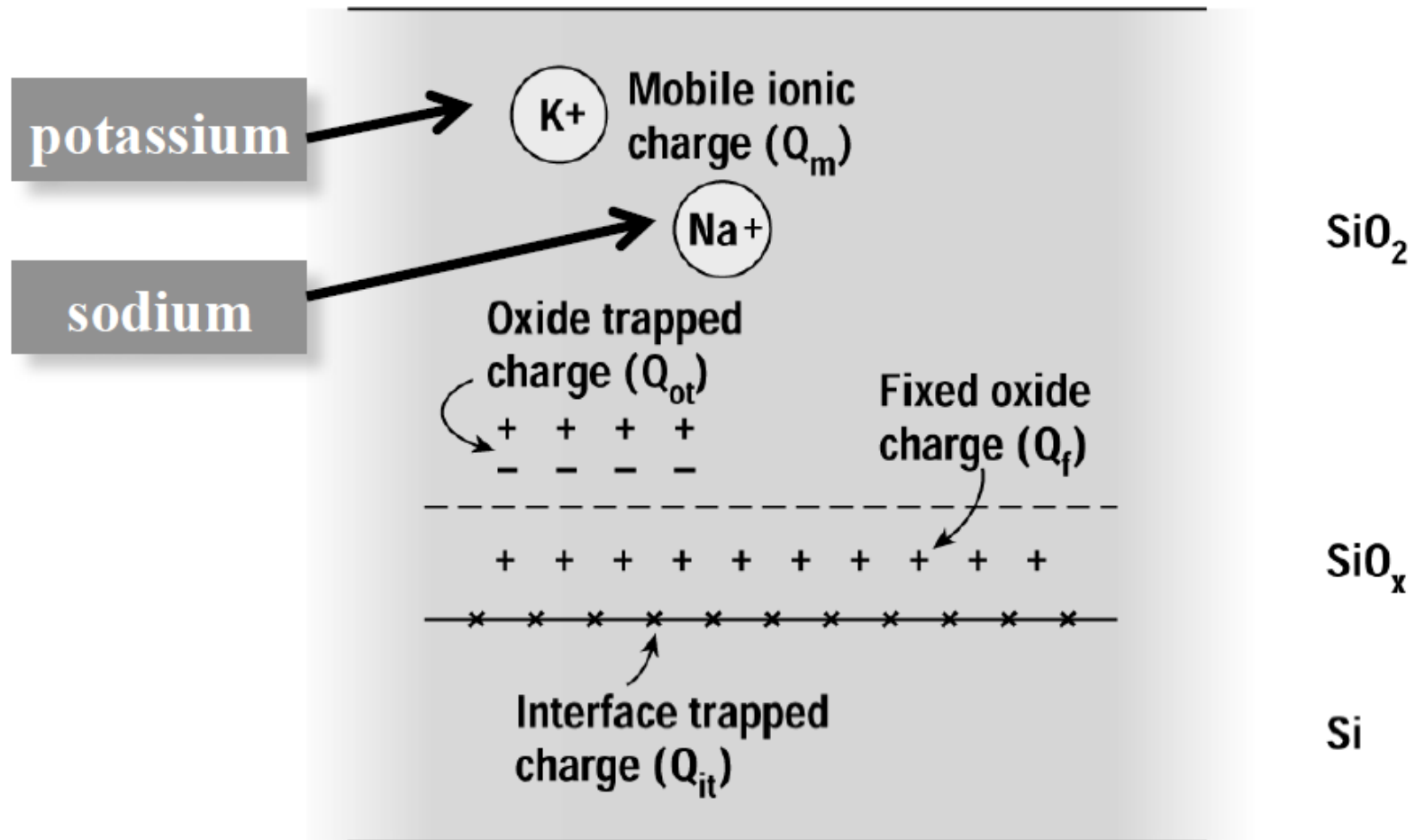
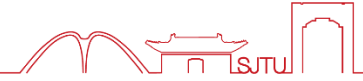


Effect of X_0 on wafer Topography



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3.4.1 Thermal oxide charges

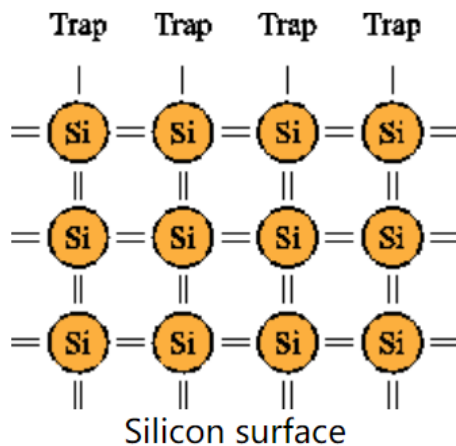


3.4.1 Oxide Quality improvement

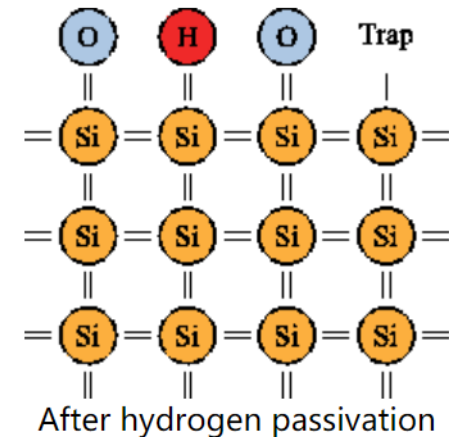
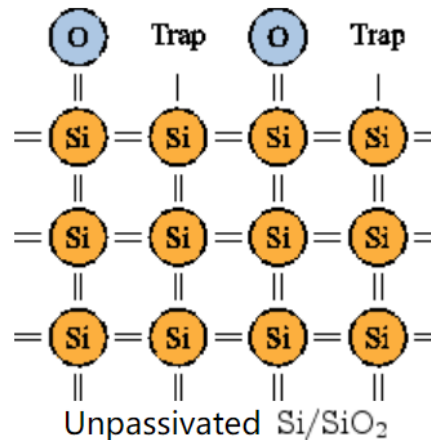


To minimize Interface Charges Q_f and Q_{it}

- Use inert gas ambient (Ar or N₂) when cooling down at end of oxidation step
- A final annealing step at 400--450°C is performed with 10%H₂+90%N₂ ambient ("*forming gas*") after the IC metallization step.



$10^{15} \text{ atom/cm}^2$



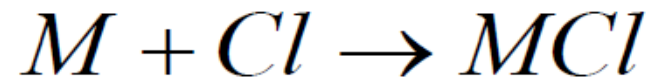
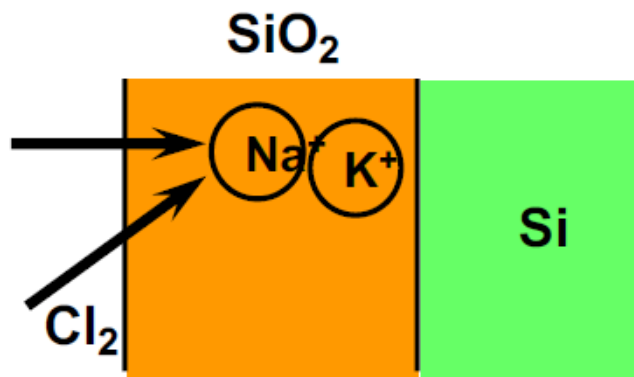
$D_{it}: 10^{10}\text{-}10^{12} \text{ atom/cm}^2$

Introduction of halogen species during oxidation

e.g. add ~1-- 5% HCl or TCE (trichloroethylene) to O₂

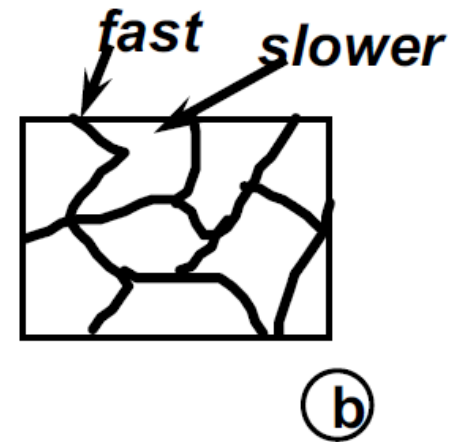
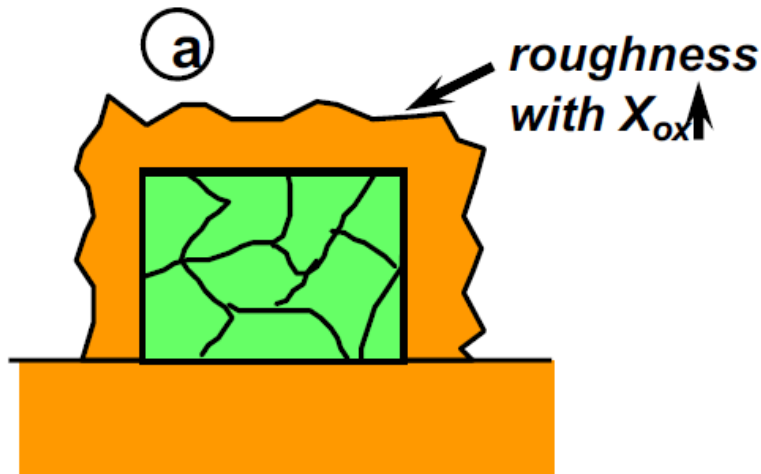
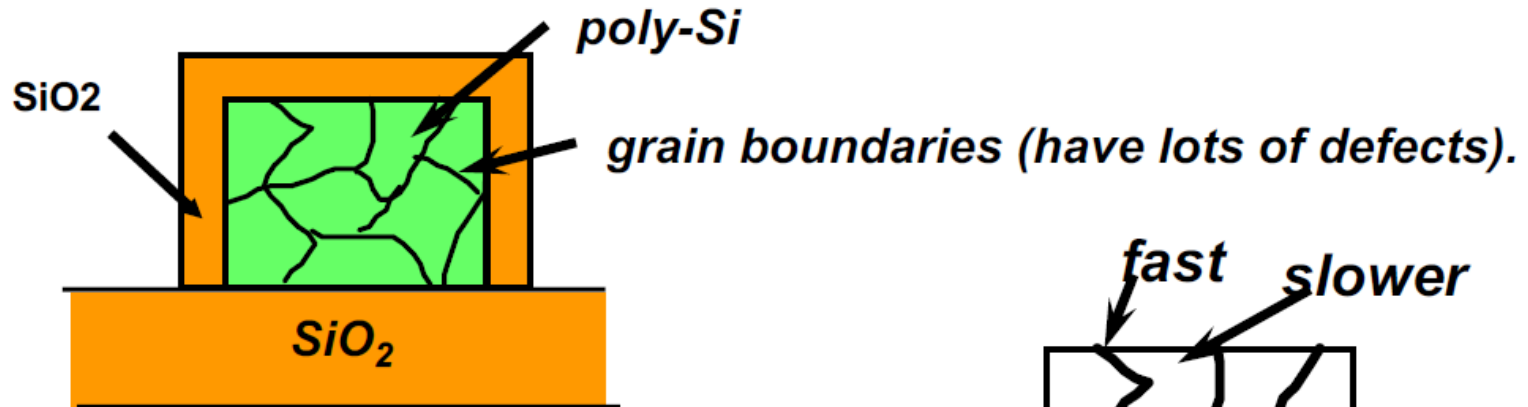
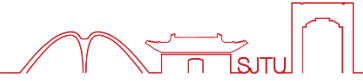
-reduction in metallic contamination

-improved SiO₂/Si interface properties



Na⁺ or K⁺ in SiO₂ are mobile!

3.4.3 Polycrystalline Si Oxidation



Overall growth rate
is higher than
single-crystal Si

- ✎ **3. 1 Why is oxidation needed?**
- ✎ **3. 2 Basics of thermal oxidation**
- ✎ **3. 3 Kinetics of thermal oxidation**
- ✎ **3. 4 Other aspects of thermal oxidation**
- ✎ **3. 5 Summary and homework**

3.5.1 Summary

➤ Why is oxidation needed? **Function and advantage for ICs**

➤ Method of thermal oxidation

Dry oxidation Wet oxidation



➤ Thermal oxidation kinetics

Deal-Grove model

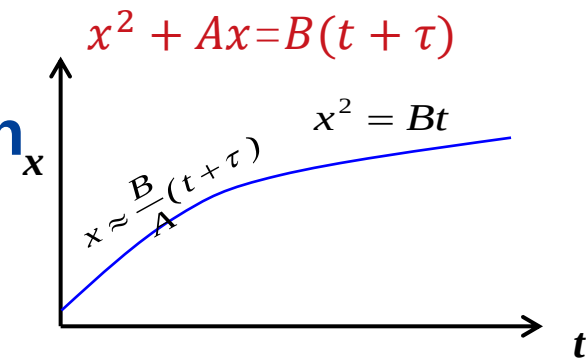
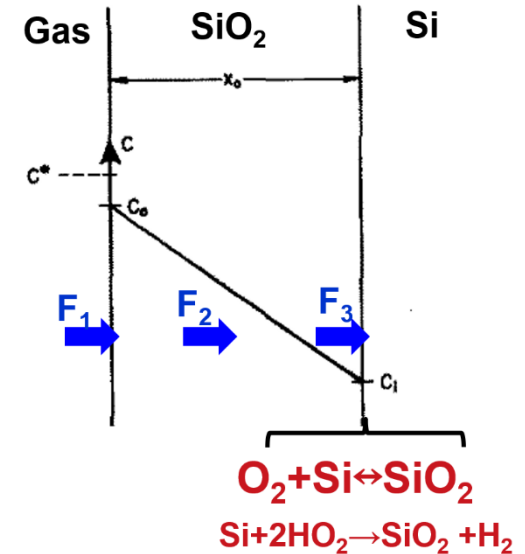
➤ Influencing factors of oxidation rate

Oxidant pressure, temperature

Surface orientation, doping concentration

Dry and wet method, initial thickness

➤ Other aspects



MANY suggestions have been made in the literature to explain the initial rapid oxidation kinetics. None have been widely accepted.

However, accurate mathematical expressions have been generated and are used in simulators (e.g. SUPREM IV and Sprocess), based upon some of these models.

Here, we will consider three of the physical models:

1. Reisman et al., Model
2. Han and Helms Model
3. Massoud et al., Model

搜索文献了解其中一种model，并用中文简述其核心内容

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谢 谢

